Smart Metro Ticket Booking System
REQUIREMENT ANALYSIS PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	

5. Data Flow Diagrams

5.1 Purpose

The data flow diagrams (DFDs) describe how information moves between the commuter, the ServiceNow-based booking system, and external services such as payment gateways and notification channels. They provide the logical foundation for the Flow Designer workflows and table design used to implement the solution.

5.2 Level 0 DFD (Context Diagram)

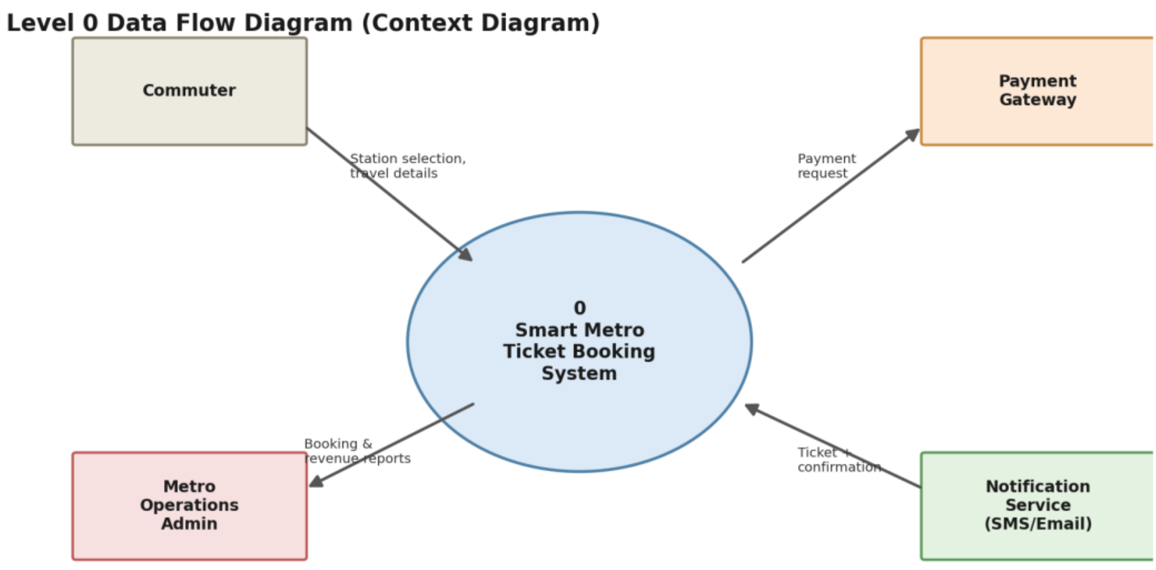

Figure 5.1: Context diagram showing the system boundary and external entities

5.3 Level 1 DFD (Process Breakdown)

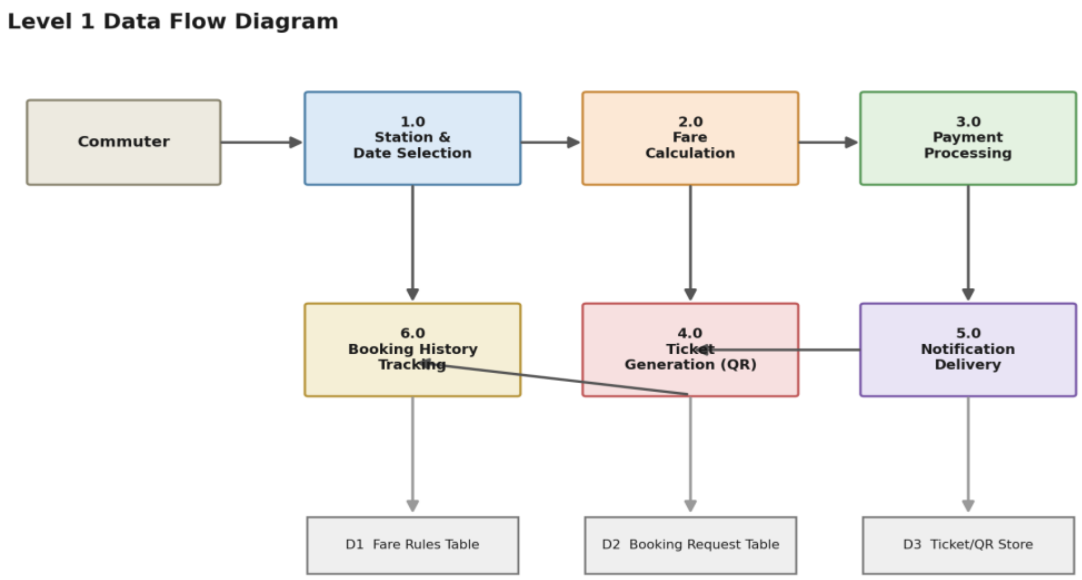

Figure 5.2: Detailed process flow from station selection to booking history tracking

5.4 Data Flow Description

Data Flow	Description
Commuter → Process 1.0	Source station, destination station, travel date, passenger count
Process 1.0 → Process 2.0	Validated station and passenger details
Process 2.0 → Process 3.0	Calculated fare amount
Process 3.0 → Process 4.0	Payment confirmation status
Process 4.0 → Process 5.0	Digital ticket with embedded QR code
Process 5.0 → Commuter	Ticket and booking confirmation via SMS/Email/Push
Process 1.0/4.0 → Process 6.0	Booking request and ticket details for history tracking
Process 2.0 ↔ D1	Fare rules and zone/distance-based pricing lookup
Process 1.0/6.0 ↔ D2	Booking request record creation and retrieval
Process 4.0 ↔ D3	Ticket and QR code storage

6. User Stories

6.1 User Story Format

Each user story follows the standard agile format: *"As a <role>, I want <goal>, so that <benefit>."* Stories are prioritized using the same P0–P3 scale defined in the Brainstorming and Idea Prioritization document.

6.2 User Stories

ID	Role	I want to…	So that…	Priorit y
US-01	Commuter	select source and destination stations and travel date from a catalog form	I can book a ticket without visiting the counter	P0
US-02	Commuter	see the fare calculated automatically before confirming	I know the exact cost upfront with no manual errors	P0
US-03	Commuter	receive a QR-code digital ticket instantly after booking	I don't need a physical token to enter the station	P0
US-04	Commuter	get an SMS/Email/Push notification with my ticket	I have proof of booking even if the app is closed	P0
US-05	Commuter	pay for my ticket using UPI, card, or wallet	I can complete payment quickly without cash	P1
US-06	Commuter	cancel a booking and receive a refund	I'm not charged for a trip I no longer need	P1
US-07	Metro Admin	view a dashboard of bookings, revenue, and station traffic	I can monitor operations and plan resources	P1
US-08	Commuter	book a season/monthly pass with renewal reminders	I don't have to book individually every day	P2
US-09	Commuter	interact with a chatbot for booking help and FAQs	I get quick answers without contacting support	P2
US-10	Elderly/Differently -abled Commuter	use a voice-assisted accessibility mode	I can book a ticket independently and confidently	P2

6.3 Traceability to MVP Scope

User stories US-01 to US-04 map directly to the P0 (Must Have) ideas finalized in the Brainstorming and Idea Prioritization document, and form the core Phase 1 MVP: catalog-based booking, automated fare calculation, QR-code ticket generation, and instant notification delivery.